MEMORANDUM

To: Newcastle Planning Board **From:** Ben Frey *(Article 3 code-amendment drafter)* **Date:** May 30, 2026 **Re:** Reducing right-of-way (ROW) width minimums in the draft Article 3 Street/Road Type Standards (Tables 3.1a & 3.1b) — justification and proposed widths **Status: Discussion draft.** The Moderate option recommended in §2 has since been written into the draft code (draft v0.4.4: Tables 3.1a/3.1b and the new Article 3 §3.d), and the Comprehensive Plan citations in §3.6 have now been filled in. This memo is retained as the supporting justification; it has not been formally adopted, and its numbers remain open to revision by the Board.

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1. The issue

Four of the ten Street/Road Types in the draft Article 3 carry a **fixed 50-foot right-of-way minimum**:

- **S-2 Village Street**
- **S-3 Neighborhood Street**
- **R-1 Connector Road**
- **R-2 Rural Road**

Experience administering the code indicates the 50-foot ROW is wider than these contexts require for most applications, and that the requirement imposes avoidable burdens — land taken from buildable/taxable parcels, additional impervious surface, and pavement and drainage the Town must maintain in perpetuity — without a corresponding safety or function benefit. This memo assembles the justification for reducing those minimums and proposes calibrated widths.

The recommendation is consistent with the central premise of the Core Zoning Code itself: that public-realm standards should be **calibrated to context (the transect)** rather than applied as a single blanket number. A uniform 50-foot ROW across a Village Street and a Rural Road is precisely the context-blind, ownership-era standard the form-based Article 3 was written to replace.

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2. Summary recommendation (Moderate option)

Convert the four fixed 50-foot minimums into **calibrated ranges with a 40-foot floor**, so that 50 feet becomes an available *ceiling* rather than a *mandate*:

Type	Current ROW	**Proposed ROW**	Change to the required minimum
S-2 Village Street	50 ft (fixed)	**40–54 ft**	−10 ft floor
S-3 Neighborhood Street	50 ft (fixed)	**40–46 ft**	−10 ft floor
R-1 Connector Road	50 ft (fixed)	**40–50 ft**	−10 ft floor
R-2 Rural Road	50 ft (fixed)	**40–50 ft**	−10 ft floor
S-1, S-4, S-5, R-3	(already ranges)	unchanged	—

The travel lanes, sidewalks, planting strips, shoulders, and ditches are not reduced. The ~20% reduction in the required ROW comes entirely out of the *unallocated margin* between the built cross-section and the ROW line.

Why **40 feet** as the common floor:

- It still holds a **complete cross-section**: e.g., 20 ft of travel + a planting strip + a sidewalk + utility/snow/grading zones on both sides (urban), or 20 ft of travel + shoulders + two full drainage ditches (rural).
- It sits **between the historic 2-rod (33 ft) and 3-rod (49.5 ft)** survey widths — i.e., it remains rooted in the same New England rod tradition the rest of the table already uses (see §3.1), simply declining to default to the widest.
- It is a **round, administrable** number that staff and applicants can apply without case-by-case derivation.

(Alternative postures — a more aggressive reduction, or fixing only S-2 — are set out in §6.)

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3. Justification

3.1 The 50-foot standard is a historical artifact, not an engineering derivation

Early New England town ways were laid out **by the rod** (1 rod = 16.5 ft): inter-town/county roads at **4 rods = 66 ft**, ordinary town roads at **3 rods = 49.5 ft**, and minor ways at **2 rods = 33 ft**. The familiar “50-foot” minimum is simply **3 rods, rounded up** — a 19th-century petition-and-survey convention, not a figure derived from what a modern street carries.

The draft code is *already* built on this tradition: **S-1 Main Street’s 66-foot ROW is 4 rods**, and **R-3 Rural Lane’s 33-foot floor is 2 rods**. The four fixed-50 Types are merely defaulting to the 3-rod middle. Nothing about the number reflects a calculated need, so moving off it sacrifices no engineering basis.

3.2 No Maine statute sets a minimum ROW width — it is the Town's to set

Maine's subdivision review statute, **30-A M.R.S.A. §4404**, requires only (criterion 5) that a subdivision *"will not cause unreasonable highway or public road congestion or unsafe conditions."* It specifies **no minimum right-of-way or pavement width**. Dimensional street standards are expressly left to the municipality. Reducing the ROW minimum is therefore squarely within the Town's home-rule authority and raises no conflict with state law, provided the resulting streets remain safe and adequate (which the retained travel-way and sight-distance standards ensure — see §4).

3.3 The 50-foot figure is uncalibrated boilerplate

The identical "no street right-of-way less than 50 feet" line appears verbatim in the subdivision ordinances of **Wells, Durham, Biddeford, and Brownfield**, among others — frequently framed as a floor to be *added to* for parking or drainage. It is a lowest-common-denominator default copied from town to town, not a standard any of those communities calibrated to a specific street function. Newcastle is free to do better by tying ROW to the actual Type cross-section.

3.4 The Type cross-sections do not use the 50 feet

A component-by-component budget of each Type (travel lanes + parking + planting strips + sidewalks, with curbs/shoulders/ditches noted) shows the 50-foot ROW substantially exceeds the **maximum** infrastructure that can be built within it:

Type	Maximum built cross-section	Current ROW	Unused margin
S-3 Neighborhood Street	~37 ft	50 ft	**≈ 13 ft**
R-1 Connector Road	~30 ft (incl. shoulders)	50 ft	**≈ 20 ft**
R-2 Rural Road	~20 ft + ditch section	50 ft	**≈ 16+ ft**
S-2 Village Street	35 ft (no parking) – 51 ft (parking both sides)	50 ft	varies*

* S-2 is the one Type where the *maximum* optional build (on-street parking both sides at full width) slightly **exceeds** 50 ft — the proposed 40–54 ft range both lowers the floor and resolves that overflow. For the typical S-2 build (no parking, or one side), the required section is only ~35 ft.

The unused margin is not pure waste — part of it is the utility/snow/grading zone — but the analysis shows there is room to right-size without encroaching on any built element.

3.5 Narrower local streets are demonstrably safer

This is an affirmative safety argument, not merely a cost argument. The widely-cited Longmont, Colorado study (Swift, *Residential Street Typology and Injury Accident Frequency*) examined

~20,000 crash reports over eight years and found that **injury crashes rise exponentially as street width increases**, with ~24-foot streets the safest — because greater width invites higher speeds on local streets. The national recommended practice, **ITE/CNU, *Designing Walkable Urban Thoroughfares: A Context Sensitive Approach* (2010)**, is built on right-sizing thoroughfares to context for exactly this reason. Holding travel lanes narrow and reducing the surrounding ROW reinforces the low design speeds the Village and Neighborhood Types call for.

3.6 Narrower ROW advances Town and State policy

- **Stormwater / impervious cover.** Less ROW and pavement means less impervious surface and a lighter stormwater burden — consistent with Maine DEP objectives and reduced long-term drainage liability.
- **Housing and land.** Every foot of unused ROW is land removed from a buildable, taxable parcel. Right-sizing supports housing supply and the efficient use of land, consistent with the State's recent direction on reducing regulatory barriers to housing.
- **Municipal cost.** The Town builds, plows, patches, and eventually rebuilds every foot of accepted ROW. A narrower standard lowers lifecycle cost.
- **Comprehensive Plan.** The reduction directly implements the *2018 Newcastle Comprehensive Plan*, whose *Streets & Roads* chapter takes up this very question and reaches the conclusions this memo urges:
  - **A range of context-calibrated street types, not one uniform width.** The Plan finds that "Newcastle's street standards are uniform and do not take into consideration whether the street is in a pedestrian-scaled neighborhood or if it serves more as a rural conduit," and directs that "new zoning will incorporate a range of street types and standards to accommodate local conditions and desired development goals." (*Road Standards = Land Use Goals*, p. 46.) The Article 3 typology and these ROW ranges are that mechanism.
  - **"Right-sized infrastructure" and lower lifecycle cost.** The Plan asks whether the Town provides "the right-sized infrastructure for the community's needs," calls anticipated road costs "one of Newcastle's greatest financial challenges," and urges the Town to "adopt new road standards" and "reduce its exposure to future costs." (*Less road*, p. 43.)
  - **Narrower streets are safer.** A connected network "enables individual streets to become narrower, which then slows traffic and increases vehicular and pedestrian safety." (*Connectivity*, p. 44.) This is the Plan's own statement of the §3.5 safety argument.
  - **Walkability and on-street parking.** The Plan calls for streets "designed for a downtown Main Street condition, with a design speed no greater than 20-25 mph" (*Claim Main Street*, p. 74) and for "repurposing existing asphalt from too-wide travel lanes to create additional [on-street] parking" (*Modern Parking Standards*, p. 45) — supporting the S-2 parking-on-one-side calibration.
  - **Rural character and efficient land use.** Protection of rural character is "an ongoing concern of citizens," and the Plan's growth framework focuses development where "infrastructure already exists to support new growth." (*Natural & Built Landscape*, p. 116; *Make Rural Work*, pp. 100–101.)

The Plan's Regulatory Flowchart (p. 177) designates the Character-Based Code as the implementing tool for these policies. Page references are to the June 2018 plan and should be confirmed against the adopted text before relying on them at a public hearing.

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4. What the reduction does not change (safety floors retained)

The proposal deliberately leaves the safety-critical dimensions untouched, so the reduced ROW does not compromise function:

- **Travel-way widths are unchanged.** S-2/S-3/R-1 keep their 20-foot travel ways, which meet the **20-foot minimum unobstructed fire-apparatus access width** under the International Fire Code (§503). Fire access is not affected.
- **Planting strips and shoulders are retained** — the zone where dry utilities are placed and where plowed snow is stored in a Maine winter.
- **Rural ditches and backslopes are retained** — R-1 and R-2 keep full open-drainage sections within the 40-foot floor.
- **Sight distances and intersection geometry** (Tables 3.2 and the curb-return standards) are unchanged.

In short, the proposal trims the *margin*, not the *street*.

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5. Effect on existing roads

Reducing the **minimum** does not make any existing street nonconforming. Roads already laid out at 50 feet (or wider) remain fully conforming; wider-than-standard ROW is never a defect. The change relieves **new** streets and dedications going forward. Existing ROWs are governed by §13 (Nonconforming Streets & Roads) exactly as today.

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6. Options for the Board

	Floor	S-2	S-3	R-1	R-2	Trade-off
A. Moderate *(recommended)*	40 ft	40–54	40–46	40–50	40–50	Keeps full utility/snow/ditch margins; most defensible.
B. Aggressive	bare section	~38	~36	~36	**33** (2-rod)	Pushes dry utilities into easements, accepts minimal ditches; maximizes land savings but thinner margins.
C. Fix S-2 only	—	40–54	50	50	50	Resolves only the parking-overflow inconsistency; leaves the bur-

7. Recommended next steps before adoption

To complete the record and refine the numbers, staff/Board should:

1. **Pull the State's *Model Subdivision Regulations for Use by Maine Planning Boards* ROW table** and confirm how it tiers width by street class (the model is the baseline most Maine ordinances copied).
2. **Compare neighboring towns** — Damariscotta, Wiscasset, Edgecomb, Bristol — for any already using tiered or context-based ROW, to show Newcastle is within (or leading) regional practice.
3. **Get Public Works / Road Commissioner input** on snow-storage and plowing needs and on dry-utility placement at a 40-foot section, to validate the retained margins.
4. **Confirm the Comprehensive Plan citations** now incorporated in §3.6 above and in Article 3 §3.d against the adopted June 2018 plan text before relying on them at hearing.
5. **On the Board's final decision, confirm or revise the widths in the draft.** The recommended Moderate option has already been written into draft v0.4.4 (Tables 3.1a/3.1b and the §3.d "Basis for Right-of-Way Widths" note) so the Board can review it in context; if the Board settles on a different option, update those tables and the §3.d note to match and regenerate the redline and Summary of Changes.

8. Sources

- 30-A M.R.S.A. §4404, Review criteria (criterion 5) — Maine Legislature.
- ITE/CNU, *Designing Walkable Urban Thoroughfares: A Context Sensitive Approach* (2010) — ITE Recommended Practice (via FHWA / Congress for the New Urbanism).
- W. Swift et al., *Residential Street Typology and Injury Accident Frequency* (Longmont, CO) — summarized by the Congress for the New Urbanism, "Narrow streets are the safest."
- New England rod-based road layout (4 rods = 66 ft; 3 rods = 49.5 ft; 2 rods = 33 ft) — National Association of Counties, "Four Rod Roads and GIS"; Vermont Agency of Transportation, *The History and Law of Vermont Town Roads.*
- Comparative Maine "50-foot" subdivision ROW boilerplate — Towns of Wells, Durham, Brownfield and City of Biddeford subdivision/street ordinances.
- International Fire Code §503 — 20-foot minimum fire-apparatus access road width.
- *Model Subdivision Regulations for Use by Maine Planning Boards* — Maine DACF Municipal Planning Assistance (recommended follow-up, §7.1).

Prepared as a discussion document for the Newcastle Planning Board. Citations are provided for verification; the drafter recommends the Board confirm each against the primary source before relying on it in a public hearing.